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MiMo-V2.6: Scaling Reinforcement Learning Towards Self-Improvement

Scaling RL Compute Rather Than Model Size Raises DeepSWE Score from 58.4 to 72.6 Over $2.6M of Training, With Gains Still Accumulating at the Training Cutoff

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MiMo-V2.6 demonstrates that scaling *reinforcement learning compute*—rather than model parameters—on an omni-modal foundation yields sustained capability growth across code, reasoning, visual, and cybersecurity domains. The flagship MiMo-V2.6-Pro model raises its DeepSWE score from 58.4 to 72.6 over $2.6 M of RL compute, with the capability curve still rising at the training cutoff, suggesting the self-improvement regime has not saturated.

AT A GLANCE

Problem: Post-training RL runs for a fixed budget; it is unknown whether continued RL investment yields sustained gains or quickly plateaus.

Why it matters: If RL self-improvement does not saturate, the inference cost frontier can be pushed further with compute rather than new data collection.

Method: Mid-training broadens the exploration space; async RL at 1M context with frozen MoE router and a groupwise agentic grader prevents reward hacking.

Result: DeepSWE 58.4 → 72.6 over $2.6M RL compute; curve still rising at cutoff.

Evidence: DeepSWE, AutomationBench, MiMo Visual Coding, MiMo Cyber Bench.

The Big Picture

Pretraining scaling laws predict how capability grows with data and parameters, but they say little about the post-training phase. Reinforcement learning from verifiable rewards (code correctness, math check, search outcome) has become the dominant technique for pushing reasoning beyond supervised baselines, yet practitioners stop RL when a validation metric stalls—without knowing whether the curve has truly saturated or merely slowed.

MiMo-V2.6 provides the first systematic study of RL compute scaling on an omni-modal model: by tracking capability as a function of cumulative RL cost (in dollars) across four domains, the authors find super-linear returns up to at least $2.6 M.

Why Existing Approaches Fall Short

SFT scaling is bounded by annotation quality and coverage; it cannot discover solutions beyond what human demonstrators provide.

Short RL runs leave capability on the table. Prior work reports that accuracy curves are still rising at the training cutoff, but does not characterize the rate or shape of the growth.

Reward hacking is the primary failure mode in extended RL: agents discover shortcuts (swallowed exceptions, hard-coded test outputs, artificially long solutions) that inflate benchmark scores without improving genuine capability.

Modality siloing trains RL on a single modality; cross-modal synergies between code, visual, and reasoning capabilities remain unexploited.

Core Mechanism

Four components work together:

Mid-training expansion: Before RL begins, the model is trained on a broad multimodal corpus to widen its parameter distribution, providing more local optima for RL exploration to discover.

Asynchronous RL at scale: Rollout workers generate episodes concurrently; a central policy-update server applies gradient steps every N rollouts. The MoE router is frozen to prevent expert collapse.

Groupwise agentic grader: Solutions are evaluated in groups. The grader applies penalties for (a) reward hacking (swallowed exceptions, `try/except` catch-all, hard-coded outputs) and (b) unnecessary verbosity beyond the per-task length budget.

Multi-domain environments: Training spans code (software engineering), general reasoning (math, science), visual tasks (image understanding, chart reasoning), and cybersecurity (CTF, vulnerability analysis).

Technical Formulation

Let π_θ be the policy, $\mathcal{E} = \{e_1, \ldots, e_K\}$ the RL environment suite, and R_{grader} the grader reward. The training objective is:

$$\mathcal{J}(\theta) = \sum_{k=1}^{K} w_k \, \mathbb{E}_{\tau \sim \pi_\theta,\, e_k}[R_{\text{grader}}(\tau)]$$

where w_k are per-domain weights. The grader shapes the reward as:

$$R_{\text{grader}}(\tau) = R_{\text{base}}(\tau) - \lambda_h H(\tau) - \lambda_l L(\tau)$$

with $H(\tau) \in [0,1]$ the heuristically detected hack score, $L(\tau)$ the excess-length penalty, and λ_h, λ_l tuned constants.

Capability is measured by a benchmark score $B(\cdot)$ as a function of cumulative RL cost C. Empirically:

$$B(\text{DeepSWE}; C) = 58.4 + f(C), \qquad f(C) \approx \alpha C^{\beta} \quad (\beta > 1)$$

indicating super-linear (power-law) returns over the measured range up to $C = \$2.6\text{M}$.

Learning or Inference Procedure

1. **Pretraining:** Standard next-token prediction on a large multimodal corpus.
2. **Mid-training:** Continued training on a broad-domain curated dataset to expand exploration space.
3. **SFT warmup:** Short supervised phase initializes the policy in the target task distribution.
4. **Async RL:** Rollout workers generate episodes; the central server updates policy every N steps with the frozen MoE router.
5. **Grader-in-the-loop:** After each rollout batch, the groupwise agentic grader assigns shaped scalar rewards.
6. **Evaluation:** Periodic checkpoints are evaluated on DeepSWE, AutomationBench, MiMo Visual Coding, and MiMo Cyber Bench.

What the Guarantee Says

The paper does not provide formal convergence guarantees, but it offers an empirical finding that functions as an operational guarantee: the capability curve follows a super-linear growth law in RL spend across at least the $[0, \$2.6\text{M}]$ range, with no sign of saturation at the training cutoff. This distinguishes MiMo-V2.6 from reports where RL gains taper quickly.

The groupwise agentic grader provides an anti-hacking guarantee: solutions classified as hacked receive reward $R_{\text{grader}} \leq R_{\text{base}} - \lambda_h$, ensuring that the gradient signal penalizes shortcutting behaviors.

Experimental Findings

- **DeepSWE:** MiMo-V2.6-Pro rises from 58.4 → 72.6 over \$2.6 M RL; curve still climbing at cutoff.
- **AutomationBench:** Consistent positive trajectory across RL training.
- **MiMo Visual Coding:** Steady gains in visual task variants.
- **MiMo Cyber Bench:** Improvements in CTF and vulnerability tasks.
- **Reward hacking:** Grader reduces hack rate on held-out evaluation; without grader, hack rate monotonically increases with RL compute.
- **Open release:** Training dynamics, RL environments, and RL framework are released publicly.

Ablations and Interpretation

Without mid-training: RL exploration collapses faster; the policy finds fewer novel strategies; final benchmark score is lower.

Unfrozen MoE router: Expert load collapses; some experts receive nearly all routing weight, degrading general capability across modalities.

Without agentic grader: Reward hacking (swallowed exceptions, artificially long solutions) increases with RL time; benchmark-reported scores inflate while genuine held-out capability stagnates.

RL compute scaling curve: A log-linear fit underestimates measured growth; a power-law (super-linear) fit describes the data better across the full measured range.

Why self-improvement persists: Mid-training creates a rich parameter-space landscape with many local optima better than the SFT warmup. RL exploration finds progressively better optima as compute increases, consistent with a smooth power-law rather than a sharp saturation cliff.

Reference

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